Lab No: 06

✵ Lab Title: For Loop in C++

○ Learning objectives:

The following topics will be covered in this lab session

- For Loop
- Nested For Loop

➪ For Loop:

A for loop is a repetition control structure that allows you to efficiently write a loop that needs to execute a specific number of times.

Syntax:

The syntax of a for loop in C++ is:

```
for ( init; condition; increment )
{
statement(s);
}
```

➪ Here is the flow of control in a for loop:

- The **init** step is executed first, and only once. This step allows you to declare and initialize any loop control variables. You are not required to put a statement here, as long as a semicolon appears.
- Next, the **condition** is evaluated. If it is true, the body of the loop is executed. If it is false, the body of the loop does not execute and flow of control jumps to the next statement just after the for loop.
- After the body of the for loop executes, the flow of control jumps back up to the **increment** statement. This statement allows you to update any loop control variables. This statement can be left blank, as long as a semicolon appears after the condition.
- The condition is now evaluated again. If it is true, the loop executes and the process repeats itself (body of loop, then increment step, and then again condition). After the condition becomes false, the for loop terminates.

Example:

Write a program using for loop which prints the values from 10 to 1.

```
//This program prints 10 to 1 using for  loop

#include <iostream.h>
#include <conio.h>

void main()
{
        clrscr();
        for (inti=10;i>0;i--)
        {
                cout<<i<<"\n";
        }

        cout<<"outside loop now";
        getch();

}
```

Example:

Write a program that checks if a number is prime or not.

```
//checks if a number is prime
#include <iostream.h>
#include <conio.h>
void main()
{
        clrscr();
        intnum;
        cout<<"Enter a number please\n";
        cin>>num;
        int flag=0;
        for (inti=2;i<num;i++)
        {
                if (num%i==0)
                {
                        flag=1;
                        break;
                }
        }
```

```
        if (flag==1)
        {
                cout<<"it is not a prime number\n";
        }
        else
        {
                cout<<"it is a prime number\n";
        }
        getch();
}
}
```

Nested For Loop:

A For loop inside another For loop is called nested For loop

```
//Author: Umair khan
//Dated: 22/06/2024
//Prints the pattern

#include <iostream.h>
#include <conio.h>

void main()
{
        clrscr();


        for (intnum=7;num>0;num--)
        {
```

```
for (int j=1;j<=num;j++)
//nested loop
                {
                        cout<<j<<"\t";
                }
                cout<<""\n;

        }

        cout<<"outside loop now";
        getch();

}
```

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Lab tasks:

Task :1

. ***for*** Loop

• Write a C++ program that uses a while loop to display numbers from 1 to

10.

• Display each number on a separate line.

Program:

```
#include <iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 10; ++i) {
        std::cout << i << endl;
    }
    return 0;
}
```

- **Output:**

```
1
2
3
4
5
6
7
8
9
10
```

Task 2:

Nested ***for*** Loop : Multiplication Table

• Write a C++ program that generates a multiplication table from 1 to 10.

• Use nested ***for*** loops to iterate through the rows and columns of the table.

• Display the multiplication table in a formatted manner.

```
        #include <iostream>
using namespace std;

int main()
{
    // outer loop for rows
    for (int i = 1; i <= 10; i++)
    {
        // inner loop for columns
        for (int j = 1; j <= 10; j++)
        {
            cout << i << "x" << j << "=" << i * j
<< "\t";
        }
        cout << endl; // next line after each row
    }

    return 0;
}
```

Output:

1x1=1 1x2=2 1x3=3 ... 1x10=10
2x1=2 2x2=4 2x3=6 ... 2x10=20
3x1=3 3x2=6 3x3=9 ... 3x10=30
...
10x1=10 10x2=20 ... 10x10=100

- **Task 3: Sum using For Loop:**

    1. Write a C program that prompts the user to enter a positive integer.
    2. Use a for loop to calculate the sum of all numbers from 1 to the entered integer.
    3. Display the calculated sum.

Input:

```
#include <iostream>
using namespace std;
int main() {
    int num;
    cout << "Enter a positive integer: ";
    cin >> num;
    if (num <= 0) {
        cout << "Please enter a positive integer." << endl;
        return 1;
    }
    int sum = 0;
    for (int i = 1; i <= num; ++i) {
        sum += i;
    }
cout << "The sum of all numbers from 1 to " << num << " is "
<< sum << "." << endl;

    return 0;
}
```

Output:

```
Enter a positive integer: 23
The sum of all numbers from 1 to 23 is 276.
```

- **Task 4: Nested For Loop - Pattern Printing**

1. Write a C program that uses nested for loops to print a pattern of asterisks.
2. The pattern should form a triangle shape, with each row having an increasing number of asterisks.
3. The number of rows in the pattern should be determined by user input.

Input:

```
#include<iostream>
using namespace std;
int main ()
{
        int n;
        cout<<"Enter the number of rows: "<<endl;
        cin>>n;
        for(int i=1; i<=n; i++)
        {
        for(int j=1; j<=i; j++)
        {
        cout<<'*';
        }
        cout<<endl;
        }
        return 0;
}
```

Output:

```
Enter the number of rows:
10
*
**
***
****
*****
******
*******
********
*********
**********
```

Conclusion

In this lab, we learned:

- How to use **for loops** and **while loops**
- Implementation of **nested loops**
- Generating tables, sums, and patterns using loops

These concepts are fundamental and widely used in programming.